4. kinematics

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we have already defined some relativistic quantities.

1 1) Four vector (a^μ) :–

It has 4 components and transform like position vector (x^μ) under lorentz Transformation

$$x^\mu = (x^0,\ x^1,\ x^2,\ x^3)$$

Where

$$\left.\begin{aligned} \bar{x}^\mu &= \Lambda^\mu{}_\nu\, x^\nu \\ \text{So}\quad \bar{a}^\mu &= \Lambda^\mu{}_\nu\, a^\nu \end{aligned}\right\}$$ Lorentz transformations of any four vector a^μ and position 4-vector x^μ.

2 2) Displacement 4-vector :–

We can define $\Delta x^\mu = x^\mu_A - x^\mu_B$ as displacement 4-vector.

$\uparrow$ Four-vector of event A $\uparrow$ position 4-vector of event B.

It also transform like those coordinates under LT.

$$\left\{\ \overline{\Delta x^\mu} = \Lambda^\mu{}_\nu\, \Delta x^\nu\ \right\}$$

3 3) Proper velocity 4-vector (η^μ) & proper velocity $(\vec{\eta})$:–

To have a vector we need to devide Δx^μ by a scalar

here $\Delta\tau$ is invarient under different frames so.

we define

$$\eta^\mu = \frac{\Delta x^\mu}{\Delta\tau} \qquad \text{or} \qquad \eta^\mu = \frac{dx^\mu}{d\tau}$$

here η^μ is 4-velocity vector. It should also transform like coordinates under L.T

$$\bar{\eta}^\mu = \Lambda^\mu{}_\nu\, \eta^\nu$$

$d\tau =$ proper time

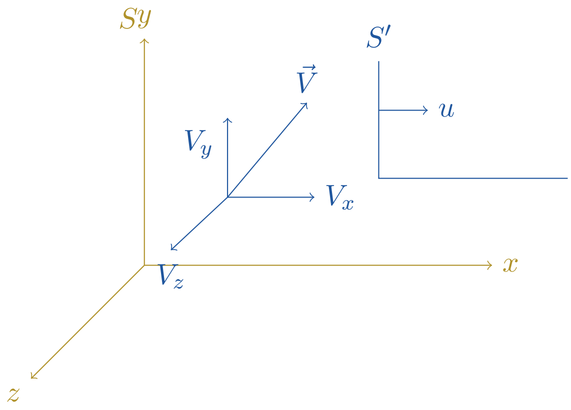
$d\tau = dt\sqrt{1-u^2/c^2}$

$$\left(\text{Here } \vec{\eta} = \text{spatial part of } \eta^\mu = \frac{m_0\vec{u}}{\sqrt{1-u^2/c^2}} = \text{proper velocity (3-D vector)}\right)$$

Remember, all 4-vectors should by def^n transform like coordinates under Lorentz transformations (change of frames).

Where as, the ordinary velocity vector should transform in different way from one frame to another

Transformation of ordinary velocity vector :–



let us study motion of a particle travelling with velocity $\vec{V}$ in frame S

Find $\vec{V}$ in frame S'.

$$\left.\vec{V}\right)_S = \vec{V} = (V_x,\ V_y,\ V_z)$$

$$\left.\vec{V}\right)_{S'} = V' = (V'_x,\ V'_y,\ V'_z)$$

$$V'_x = \frac{\Delta x'}{\Delta t'} = \frac{(\Delta x - u\Delta t)\,\gamma}{\left(\Delta t - \frac{u\Delta x}{c^2}\right)\gamma} = \frac{\left(\frac{\Delta x}{\Delta t} - u\right)}{\left(1 - \frac{u}{c^2}\frac{\Delta x}{\Delta t}\right)} = \frac{V_x - u}{1 - \frac{uV_x}{c^2}}$$

$$V'_y = \frac{\Delta y'}{\Delta t'} = \frac{\Delta y}{\left(\Delta t - \frac{u\Delta x}{c^2}\right)\gamma} = \frac{\frac{\Delta y}{\Delta t}\sqrt{1 - u^2/c^2}}{\left(1 - \frac{u}{c^2}\frac{\Delta x}{\Delta t}\right)} = \frac{V_y\sqrt{1-u^2/c^2}}{\left(1 - \frac{uV_x}{c^2}\right)}$$

$$V'_z = \frac{\Delta z'}{\Delta t'} = \frac{\Delta z}{\left(\Delta t - \frac{u\Delta x}{c^2}\right)\gamma} = \frac{\left(\frac{\Delta z}{\Delta t}\right)\sqrt{1 - u^2/c^2}}{\left(1 - \frac{u}{c^2}\frac{\Delta x}{\Delta t}\right)} = \frac{V_z\sqrt{1-u^2/c^2}}{\left(1 - \frac{uV_x}{c^2}\right)}$$

here 'u' is |Relative velocity| of S' w.r.t. S

4 (4) Relativistic energy and momentum :–

We define energy-momentum 4vector (or) Momentum 4-vector as

$$p^\mu = m_0\eta^\mu = m_0\frac{dx^\mu}{d\tau}$$

$$\text{here} \quad p^0 c = E \qquad \text{and} \qquad \vec{p} = \gamma m\vec{u} = \frac{m\vec{u}}{\sqrt{1-u^2/c^2}}$$

↗ Relativistic Energy of a particle of mass m_0

↗ Relativistic momentum of particle moving with ordinary velocity $\vec{u}$.

$$E = p^0 c = c\,m_0\eta^0 = c\,m_0\frac{dx^0}{d\tau} = \frac{m_0c^2}{\sqrt{1-u^2/c^2}}$$

$$\text{So,} \qquad E = \frac{m_0 c^2}{\sqrt{1-u^2/c^2}} \qquad \& \qquad \vec{p} = \frac{m_0 \vec{u}}{\sqrt{1-u^2/c^2}}$$

We can thus express momentum 4 vector p^μ as

$$p^\mu = \begin{pmatrix} p^0 \\ p^1 \\ p^2 \\ p^3 \end{pmatrix} = \begin{pmatrix} p^0 \\ \vec{p} \end{pmatrix} = \left(\frac{m_0 c}{\sqrt{1-u^2/c^2}}, \frac{m_0 \vec{u}}{\sqrt{1-u^2/c^2}} \right)$$

This 4-vector must also transform like coordinates under L.T.

$$\bar{p}^\mu = \Lambda^\mu{}_\nu \, p^\nu$$

$$\text{i.e.} \quad \left\{ \begin{array}{l} \bar{p}^0 = (p^0 - \beta p^1)\, \gamma \\ \bar{p}^1 = (p^1 - \beta p^0)\, \gamma \\ \bar{p}^2 = p^2 \\ \bar{p}^3 = p^3 \end{array} \right\} \quad \begin{array}{l} \text{For motion of frame } S' \text{ along} \\ x\, x' \ \underline{\text{axis}} \text{ with speed } u \\ \gamma = \dfrac{1}{\sqrt{1-u^2/c^2}} \end{array}$$

In every closed system Total relativistic energy and momentum are conserved.

Invarient — Same value in different frames of reference

Conserved — Same value before and after some process.

Quantity	Invarient	Conserved
mass	✓	×
Energy	×	✓
electric charge	✓	✓
Velocity	×	×
$a^\mu a_\mu$	✓	?

let us find the Invarient scalar product of 4-momentum vector

$$\begin{aligned} I &= p^\mu p_\mu = -(p^0)^2 + \vec{p}\cdot\vec{p} \\ &= -\left(\frac{mc}{\sqrt{1-u^2/c^2}} \right)^2 + \left(\frac{m\vec{u}}{\sqrt{1-u^2/c^2}} \cdot \frac{m\vec{u}}{\sqrt{1-u^2/c^2}} \right) \\ &= \frac{-m^2c^2}{1-u^2/c^2} + \frac{m^2u^2}{1-u^2/c^2} \\ &= -\frac{m^2c^2(c^2-u^2)}{(c^2-u^2)} = -m^2c^2 \end{aligned}$$

$$\text{i.e.} \qquad p^\mu = (p^0,\ \vec{p}) = \left(\frac{E}{c},\ \vec{p} \right)$$

$$p_\mu = (-p_0,\ \vec{p}) = (-E/c,\ \vec{p})$$

$$I = p^\mu p_\mu = -\frac{E^2}{c^2} + p^2 = -m^2c^2$$

$$\Rightarrow \qquad \boxed{E^2 - p^2c^2 = m^2c^4}$$

We can define idea of Invarient mass as

$$m = \frac{-p^\mu p_\mu}{c^2} \qquad \left(\begin{array}{l} \text{But this is not in general} \\ \text{Sum of Individual mass} \end{array} \right)$$

5 (R4) Invarient Quantities :–

Coordinate of any event $= X = (ct,\ x, y, z) = (x^0, x^1, x^2, x^3) = x^\mu$

$$\text{under } \underline{\text{L.T.}} \qquad \bar{x}^\mu = \Lambda^\mu{}_\nu x^\nu$$

$$\text{i.e.} \quad \begin{aligned} \bar{x}^1 &= (x^1 - \beta x^0)\,\gamma \\ \bar{x}^0 &= (x^0 - \beta x^1)\,\gamma \end{aligned}$$

we have defined the invarient

$$\begin{pmatrix} x^\mu x_\mu = X \cdot X = x^2 = -(x^0)^2 + \vec{r}\cdot\vec{r} = -(x^0)^2 + r^2 \\ \Delta x^\mu \Delta x_\mu = (\Delta x)^2 = -(\Delta x^0)^2 + (\Delta x^1)^2 + (\Delta x^2)^2 + (\Delta x^3)^2 \end{pmatrix}$$

$$\text{Similarly} \qquad p^\mu p_\mu \ = \ \bar{p}^\mu \bar{p}_\mu \qquad \text{(invarient)}$$

$$\{\text{let me prove it using tensor notation.}\} \rightarrow$$

p^μ transform like coordinates under LT

$$\{\bar{p}^\mu = \Lambda^\mu{}_\nu\, p^\nu\} \qquad \{\bar{p}_\mu = \Lambda^{-1}{}^\mu_\nu\, p_\nu\}$$

$$\Rightarrow \quad \bar{p}^\mu \bar{p}_\mu \ = \ \Lambda^\mu{}_\nu\, p^\nu\, \Lambda^{-1}{}^\mu_\nu\, p_\nu \ = \ \left(\Lambda^\mu{}_\nu \Lambda^{-1}{}^\mu_\nu\right) p^\mu p_\mu$$

$$\boxed{\bar{p}^\mu\, \bar{p}_\mu \ = \ p^\mu p_\mu}$$

$$\begin{aligned} \bar{p}^0 &= (p^0 - \beta p^1)\,\gamma \\ \bar{p}^1 &= (p^1 - \beta p^0)\,\gamma \\ \bar{p}^2 &= p^2 \\ \bar{p}^3 &= p^3 \end{aligned} \quad \Rightarrow \quad \begin{aligned} -\bar{p}_0 &= (-p_0 - \beta p_1)\,\gamma \quad \Rightarrow \quad \bar{p}_0 = (p_0 + \beta p_1)\,\gamma \\ \bar{p}_1 &= (p_1 + \beta p_0)\,\gamma \\ \bar{p}_2 &= p_2 \\ \bar{p}_3 &= p_3 \end{aligned}$$

$$\Lambda^\mu{}_\nu \ = \begin{pmatrix} \gamma & -\beta\gamma & 0 & 0 \\ -\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

We could have defined Invarient quantity $I = a^\mu b_\mu$ as in both ways.

$$\left.\begin{aligned} I &= a^\mu b_\mu = -(a^0 b^0) + \vec{a}\cdot\vec{b} \\ \text{or} \quad I &= -a^\mu b_\mu = (a^0 b^0) - \vec{a}\cdot\vec{b} \end{aligned}\right\} \quad \text{Both are correct and in use in literature.}$$

For any two four vectors A & B.

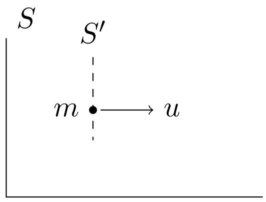
$$A^\mu = (A^0,\ \vec{A}\,)$$
$$B^\mu = (B^0,\ \vec{B}\,)$$

$$I = A^\mu B_\mu \ = \ -A^0 B^0 + \vec{A}\cdot\vec{B} \ = \ \text{Invarient}$$
$$\text{or} \quad \underline{I} = -A^\mu B_\mu \ = \ A^0 B^0 - \vec{A}\cdot\vec{B} \ = \ \text{Invarient}$$

$$\text{For special case } \vec{A} = \vec{B}, \qquad I = A^\mu A_\mu = -(A^0)^2 + |A|^2 = \text{Invarient}$$

$$\text{Similarly for } A^\mu = p^\mu \quad \Rightarrow \quad I = (p^0)^2 - p^2 = m^2 c^2$$

So Four vector of any kind has its $(\text{norm})^2 = a^\mu a_\mu$ which is Invarient under LT (under different frames it is same).



$$p^\mu\Big)_S = \left(\frac{E}{c},\ \vec{p}\right)$$

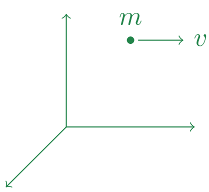
$$\bar{p}^\mu\Big)_{S'} = \left(\frac{mc}{\sqrt{1-u^2/c^2}},\ \frac{m\cancelto{0}{\vec{u}}}{\sqrt{1-u^2/c^2}}\right) \qquad \begin{cases} u = \text{velocity of} \\ \text{particle} \end{cases}$$

$$\bar{p}^\mu = (mc,\ 0)$$

$$\overline{p^\mu}\cdot\bar{p}_\mu = m^2c^2 \qquad \text{Which is Same}$$

$$p^\mu p_\mu = \frac{E^2}{c^2} - p^2 = m^2c^2$$

Shortcut for $p^\mu p_\mu = -m^2c^2$



In frame of particle

$$\begin{aligned} p &= (p^0,\ \vec{p}) \\ &= \left(\frac{mc}{\sqrt{1-\cancelto{0}{u^2}/c^2}},\ \frac{m\cancelto{0}{\vec{u}}}{\sqrt{1-u^2/c^2}}\right) \\ &= (mc,\ 0) \end{aligned}$$

$$\boxed{p\cdot p = m^2c^2}$$

For any particle of energy E & momentum $\vec{p}$

$$p = (p^0,\ \vec{p}) = \left(\frac{E}{c},\ \vec{p}\right) \qquad \begin{cases} E = \dfrac{mc^2}{\sqrt{1-u^2/c^2}} \\ \vec{p} = \dfrac{m\vec{u}}{\sqrt{1-u^2/c^2}} \end{cases}$$

$$p\cdot p = m^2c^2 = \frac{E^2}{c^2} - p^2$$

$$\Rightarrow \qquad \boxed{E^2 = p^2c^2 + m^2c^4}$$

For a photon $m_0 = 0 \quad \Rightarrow \quad \{E = pc\}$

Summary :–

$$p^\mu = \begin{pmatrix} p^0 \\ p^1 \\ p^2 \\ p^3 \end{pmatrix} = \begin{pmatrix} p^0 \\ \vec{p} \end{pmatrix} = \begin{pmatrix} mc/\sqrt{1-u^2/c^2} \\ m\vec{u}/\sqrt{1-u^2/c^2} \end{pmatrix}$$

$$\text{with} \qquad p^\mu p_\mu = -m^2c^2$$

$$\text{Where} \qquad p_\mu = (-p_0,\ \vec{p})$$

6 Questions

(Q.1) Write four vector for photon.

Ans)

$$p^\mu = (p^0,\ p^1,\ p^2,\ p^3)$$

$$p^\mu = m\,\eta^\mu = m\,\frac{dx^\mu}{d\tau} \qquad \text{where} \quad x^\mu = \begin{pmatrix} x^0 \\ x^1 \\ x^2 \\ x^3 \end{pmatrix} = \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix}$$

$$= \frac{m}{\sqrt{1-u^2/c^2}} \begin{pmatrix} c \\ \vec{u} \end{pmatrix}$$

$$= \begin{pmatrix} \dfrac{mc}{\sqrt{1-u^2/c^2}} \\ \dfrac{m\vec{u}}{\sqrt{1-u^2/c^2}} \end{pmatrix}$$

$$\text{for photon} \qquad p^\mu = \left(\frac{E}{c},\ \vec{p}\right) \qquad \frac{E^2}{c^2} - p^2 = m_0^2c^2 = 0$$

$$\Rightarrow \quad \left(\frac{E}{c} = p\right)$$

$$\left\{ \text{So} \quad p^\mu = \left(\frac{E}{c},\ \vec{p}\right) \quad \text{such that} \quad E = pc \right\}$$

(Q.1)

$$\begin{array}{ccccc} W & m & & m' & \\ \rightsquigarrow & \bullet & \longrightarrow & \circ \longrightarrow v & \text{(Find } m' \text{ \& } v\text{)} \\ \downarrow & \downarrow & & \downarrow & \\ p_1^\mu & p_2^\mu & & p_3^\mu & \end{array}$$

$$p_1^\mu + p_2^\mu = p_3^\mu$$

$$\left(\frac{E}{c},\ \vec{p}\right) + (mc,\ 0) = \left(\frac{m'c}{\sqrt{1-v^2/c^2}},\ \frac{m'v}{\sqrt{1-v^2/c^2}}\right)$$

$$\frac{E}{c} + mc = \frac{m'c}{\sqrt{1-v^2/c^2}}$$

$$\& \qquad p = \frac{m'v}{\sqrt{1-v^2/c^2}}$$

$$\text{from frame of } (m') \qquad \bar{p}_3^{\mu} = (m'c,\ 0)$$

$$\bar{p}_3^{\mu} \cdot \bar{p}_{3\mu} = m'^2c^2 = (p_1+p_2)\cdot(p_1+p_2)$$

$$\bar{p}_3^{\mu}\bar{p}_{3\mu} = m'^2c^2 = \left(\frac{E}{c}+mc,\ \vec{p}\right)\cdot\left(\frac{E}{c}+mc,\ \vec{p}\right)$$

$$m'^2c^2 = \left(\frac{E}{c}+mc\right)^2 - p^2$$

$$p = \frac{E}{c} \quad \Rightarrow \quad (m'c)^2 = \frac{E^2}{c^2} + m^2c^2 + 2mE - \frac{E^2}{c^2}$$

$$(m'c)^2 = m^2c^2 + 2mE$$

$$m' = \sqrt{m^2 + \frac{2mE}{c^2}}$$

$$\text{Where} \quad E = h\nu \quad \text{or} \quad \hbar\omega$$

$$\boxed{m' = \sqrt{m^2 + \frac{2m\hbar\omega}{c^2}}}$$

(Q.2)

$$p \underset{u}{\longrightarrow} \underset{\text{Rest}}{p} \longrightarrow p+p+p+\bar{p}$$

$\hookrightarrow$ antiproton

with how much energy first proton should be imparted so that we have $3p+\bar{p}$ moving together with some final velocity.

$$p+p \longrightarrow \{p+p+p+\bar{p}\} \longrightarrow V$$

$$\left(\frac{mc}{\sqrt{1-u^2/c^2}},\ \vec{p}\right) + (mc,\ 0) = \left(\frac{4mc}{\sqrt{1-V^2/c^2}},\ \frac{4m\vec{V}}{\sqrt{1-V^2/c^2}}\right)$$

$$\text{or} \quad \left(\frac{E}{c},\ \vec{p}\right) + (mc,\ 0) = \left(\frac{4mc}{\sqrt{1-V^2/c^2}},\ \frac{4m\vec{V}}{\sqrt{1-V^2/c^2}}\right) = p_f^{\mu}$$

$$\bar{p}_f^{\mu}\,\bar{p}_{f\mu} = p_f^{\mu}p_{f\mu} \qquad \text{(Invarient)}$$

So let me find this result from frame of C.O.M of All four particles.

$$-\bar{p}_f^{\mu}\,\bar{p}_{f\mu} = (4mc,\ 0)\begin{pmatrix}4mc\\0\end{pmatrix} = 16m^2c^2$$

$$p_f^{\mu}p_{f\mu} = \left(\frac{E}{c}+mc,\ p\right)\begin{pmatrix}\frac{E}{c}+mc\\p\end{pmatrix} = \left(\frac{E}{c}+mc\right)^2 - p^2$$

$$\Rightarrow \qquad \frac{E^2}{c^2} + m^2c^2 + 2mE - p^2 = 16m^2c^2 \qquad (1)$$

Also from first particles

$$p_1^{\mu}p_{1\mu} = I = -m^2c^2$$

$$-\frac{E^2}{c^2} + p^2 = -m^2c^2$$

$$\left(\frac{E^2}{c^2} - p^2 = m^2c^2\right)$$

eq$^\text{n}$ ① becomes,

$$2m^2c^2 + 2mE = 16m^2c^2$$

$$\boxed{E = \frac{14m^2c^2}{2m} = 7mc^2}$$